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7330 Research Proposal

- I. Project Title
 - A. Evaluating Concurrency Control Strategies in a Multi-Entity Library Circulation and Room Booking Database System
- II. Names & Email Addresses
 - A. Joshua Samuel - joshsamuel@smu.edu
 - B. Avery John - averyjohn@smu.edu
 - C. Zachary Jennings - zjjennings@smu.edu
- III. Problem Statement
 - A. Concurrency issues, such as lost updates, dirty reads, and phantom reads, can affect data accuracy and database performance. These issues are especially relevant in library systems, where multiple patrons may attempt to check out the same book copy or reserve the same room at the same time.
- IV. Research Methodology
 - A. Overview: Investigate how different concurrency control strategies and isolation levels in MySQL affect correctness, throughput, and latency in a simulated multi-user library circulation and room-booking system.
 - B. We'll begin by designing and implementing a realistic relational schema to model library circulation of books, copies, loans, and fines as well as room reservations. Synthetic data will be generated to simulate a high-volume environment with thousands of book copies, loans and reservations. We'll test multiple concurrency control strategies against a baseline. These strategies could include pessimistic locking, optimistic locking, application-level conflict detection and trigger-based enforcement. Each of the strategies will be executed under a simulated multi-user workload issuing simultaneous checkouts/reservations, and metrics will be collected and evaluated such as: number of conflicts, deadlocks, lost updates, rollback frequency, average latency, and throughput.
- V. Previous Work
 - A. Foundational work in concurrency control has established locking and transaction isolation as the main strategy to keep data consistent while many users read and write simultaneously.
 - B. Citations:
 - 1. <https://www.csd.uoc.gr/~hy460/pdf/p624-eswaran.pdf> - formalized transaction consistency and introduced a procedure called two-phase locking, which is currently the basis for how most commercial database

systems (including MySQL's engine) handle conflicting data reads and writes.

2. <https://web.stanford.edu/class/cs345d-01/rl/SQL-isolation-critique.pdf> - they investigated transaction isolation levels and identified several concurrency-related anomalies, including dirty reads and phantom reads. Their findings demonstrate the importance of understanding how isolation levels affect database consistency when multiple transactions execute simultaneously.
3. <https://dev.mysql.com/doc/refman/8.0/en/innodb-transaction-model.html> - Authoritative source on MySQL's locking mechanisms and isolation levels
4. <https://www.engineering.upenn.edu/~zives/05s/cis650/papers/opt-cc.pdf> - proposes optimistic methods as a strategy for concurrency control, which introduced an increased efficiency in reducing system throughput
5. <https://www.vldb.org/pvldb/vol8/p209-yu.pdf?ref=blog.vvsevolodovich.dey> - these researchers evaluated multiple concurrency controls methods at very high core counts and found that lock contention and timestamp allocations become severe blockage points as concurrency grows

VI. Research Plan & Schedule

A. Milestones

1. Database Development and Data Generation - October 5th
2. Establishing Baseline Performance - October 12th
3. Simulating Concurrent Database Operations - October 12th
4. Implementing Concurrency Control Strategies - October 19th
5. Workload Simulation - October 19th
6. Performance Evaluation and Comparison - October 26th

VII. List of Resources

- A. We plan to use open source tools like MySQL Workbench and Python to generate our synthetic data and perform our analysis. We will not be using any equipment that needs to be rented out, so our budget is nonexistent.